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Class: DSC640 - Data Presentation & Visualization

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Purpose: This notebook performs the tasks outlined in Milestone 1. The full assignment will join the work found within this notebook with other milestones to create a final report.

In [1]:
```
import pandas as pd

df = pd.read_excel('nationaldatabaseofchildcareprices.xlsx')
```

In [2]:
```
# read the sample data
df.head()
```

Out[2]:

	State_Name	State_Abbreviation	County_Name	County_FIPS_Code	StudyYear	UN
0	Alabama	AL	Autauga County	1001	2008	
1	Alabama	AL	Autauga County	1001	2009	
2	Alabama	AL	Autauga County	1001	2010	
3	Alabama	AL	Autauga County	1001	2011	
4	Alabama	AL	Autauga County	1001	2012	

5 rows × 227 columns

In [3]:
```
# check the data types and columns
print(df.dtypes)
print(df.columns)
```

```
State_Name               object
State_Abbreviation       object
County_Name              object
County_FIPS_Code          int64
StudyYear                 int64
                         ...
_75FCCInfant_flag       float64
_75FCCToddler           float64
_75FCCToddler_flag      float64
_75FCCPreschool         float64
_75FCCPreschool_flag    float64
Length: 227, dtype: object
Index(['State_Name', 'State_Abbreviation', 'County_Name', 'County_FIPS_Cod
e',
       'StudyYear', 'UNR_16', 'FUNR_16', 'MUNR_16', 'UNR_20to64',
       'FUNR_20to64',
       ...
       'MFCCToddler', 'MFCCToddler_flag', 'MFCCPreschool',
       'MFCCPreschool_flag', '_75FCCInfant', '_75FCCInfant_flag',
       '_75FCCToddler', '_75FCCToddler_flag', '_75FCCPreschool',
       '_75FCCPreschool_flag'],
      dtype='object', length=227)
```

In [4]:
```
# print number of columns
num_columns = df.shape[1]
print(num_columns)
```

```
227
```

In [5]:
```
# print number of rows
num_rows = df.shape[0]
print(num_rows)
```

```
34567
```

In [6]:
```
# Find columns with the same value for all rows
constant_cols = [col for col in df.columns if df[col].nunique(dropna=False)
print("Columns with the same value in all rows:", constant_cols)
print("Number of such columns:", len(constant_cols))
```

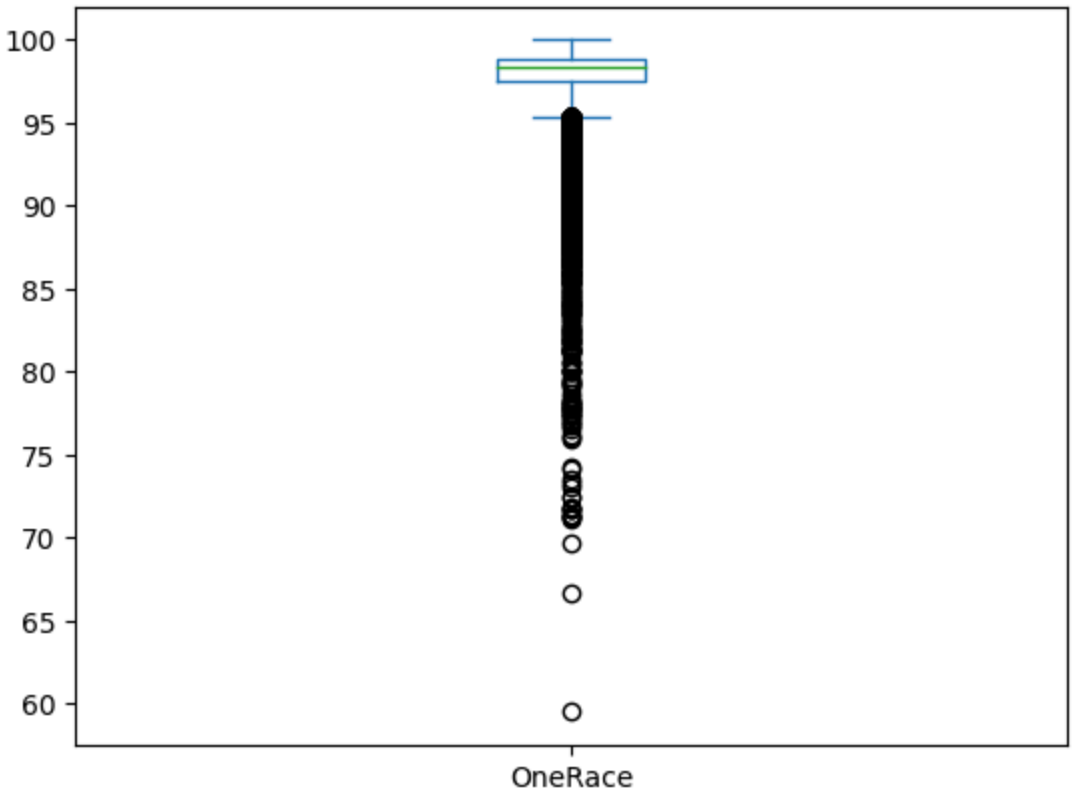
```
Columns with the same value in all rows: ['iEMP_M', 'iMEMP_M', 'iFEMP_M', 'i
EMP_Service', 'iMEMP_Service', 'iFEMP_Service', 'iEMP_Sales', 'iMEMP_Sales',
'iFEMP_Sales', 'iEMP_N', 'iMEMP_N', 'iFEMP_N', 'iEMP_P', 'iMEMP_P', 'iFEMP_
P']
Number of such columns: 15
```

In [7]:
```
# Calculate the average of the 'OneRace' column
average_one_race = df['OneRace'].mean()
print("Average of OneRace:", average_one_race)

# Create a boxplot of the 'OneRace' column
df['OneRace'].plot.box()

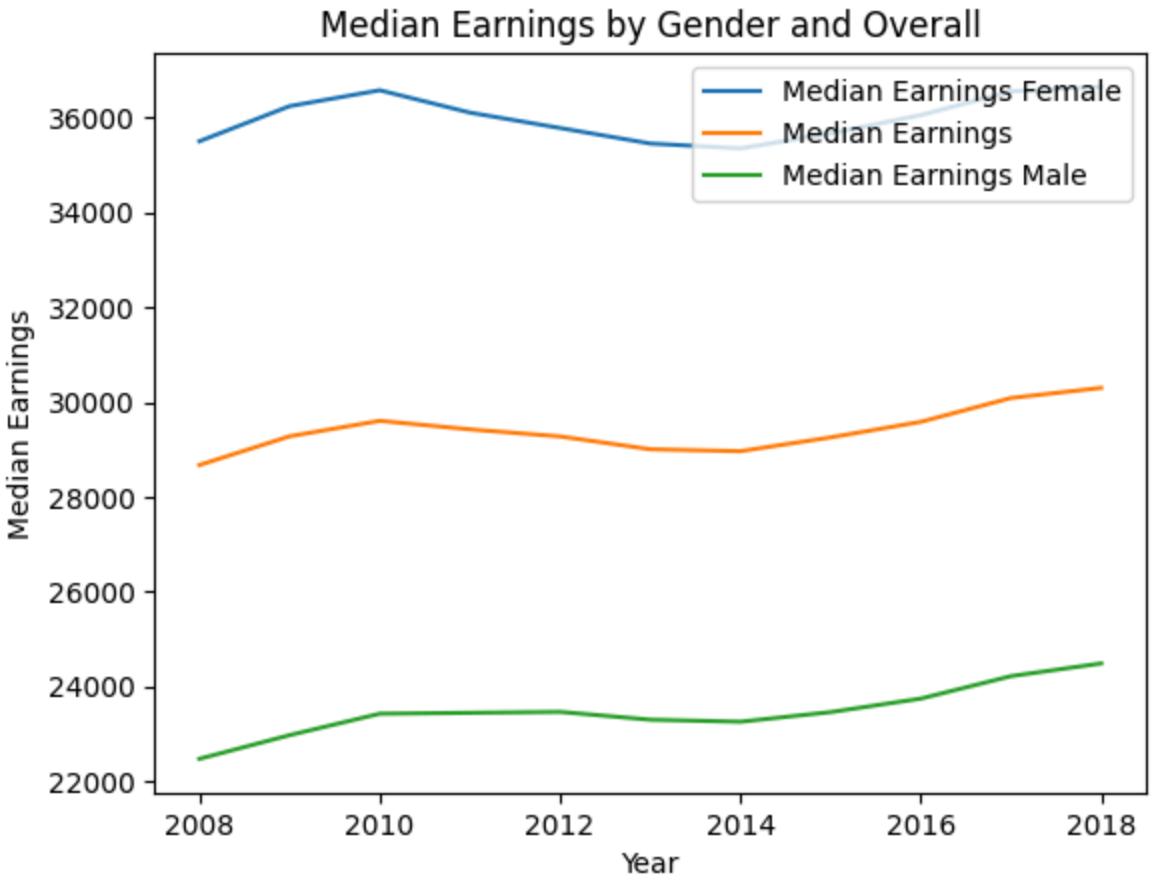
demographic_cols = ['OneRace_W','OneRace_B','OneRace_I','OneRace_A','OneRace
```

```
Average of OneRace: 97.92286284606705
```



In [8]:
```
# Plot Median Earnings Female, Median Earnings, and Median Earnings Male tog
df.groupby('StudyYear')[['MME_2018', 'ME_2018', 'FME_2018']].mean().plot(
    kind='line',
    title='Median Earnings by Gender and Overall',
    ylabel='Median Earnings',
    xlabel='Year'
).legend(['Median Earnings Female', 'Median Earnings', 'Median Earnings Male
```

Out[8]:
```
<matplotlib.legend.Legend at 0x319d29210>
```


Median Earnings by Gender and Overall
Median Earnings Female
Median Earnings
Median Earnings Male
Median Earnings
36000
34000
32000
30000
28000
26000
24000
22000
2008
2010
2012
2014
2016
2018
Year